

DATA LOADED

id.orig_p
id.resp_p
proto
service
flow_duration
fwd_pkts_tot
bwd_pkts_tot
fwd_data_pkts_tot
bwd_data_pkts_tot
fwd_pkts_per_sec
bwd_pkts_per_sec
flow_pkts_per_sec
down_up_ratio
fwd_header_size_tot
fwd_header_size_min
fwd_header_size_max
bwd_header_size_tot
bwd_header_size_min
bwd_header_size_max
flow_FIN_flag_count
flow_SYN_flag_count
flow_RST_flag_count
fwd_PSH_flag_count
bwd_PSH_flag_count
flow_ACK_flag_count
fwd_URG_flag_count
bwd_URG_flag_count
flow_CWR_flag_count
flow_ECE_flag_count
fwd_pkts_payload.min
fwd_pkts_payload.max
fwd_pkts_payload.tot
fwd_pkts_payload.avg
fwd_pkts_payload.std
bwd_pkts_payload.min
bwd_pkts_payload.max
bwd_pkts_payload.tot
bwd_pkts_payload.avg
bwd_pkts_payload.std

flow_iat_min
flow_iat_max
flow_iat_tot
flow_iat_avg
flow_iat_std
payload_bytes_per_second
fwd_subflow_pkts
bwd_subflow_pkts
fwd_subflow_bytes
bwd_subflow_bytes
fwd_bulk_bytes
bwd_bulk_bytes
fwd_bulk_packets
bwd_bulk_packets
fwd_bulk_rate
bwd_bulk_rate
active.min
active.max
active.tot
active.avg
active.std
idle.min
idle.max
idle.tot
idle.avg
idle.std
fwd_init_window_size
bwd_init_window_size
fwd_last_window_size
Attack_type

DATA VALUES:

LINE:0 [0,38667,1883,tcp,mqtt,32.011598,9,5,3,3,0.281148,0.156193,0.437341,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0,76,0,8.444444,13.11593
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LINE:1 [1,51143,1883,tcp,mqtt,31.883584000000006,9,5,3,3,0.282277,0.1568210000000002,0.439097,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0,76
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DATA VALUES:
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LINE:1 [1,51143,1883,tcp,mqtt,31.883584000000006,9,5,3,3,0.282277,0.1568210000000002,0.439097,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76
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LINE:2 [2,44761,1883,tcp,mqtt,32.124053,9,5,3,3,0.280164,0.155647,0.435811,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.74,0.8.222222,12.85279
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LINE:3 [3,60893,1883,tcp,mqtt,31.961063,9,5,3,3,0.281593,0.15644,0.438033,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.74,0.8.222222,12.852799
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LINE:4 [4,51087,1883,tcp,mqtt,31.902362,9,5,3,3,0.282111,0.156728,0.438839,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.11593
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LINE:5 [5,48579,1883,tcp,mqtt,31.869686,9,5,3,3,0.2824,0.156889,0.439289,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.1159359
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LINE:6 [6,54063,1883,tcp,mqtt,32.094711,9,5,3,3,0.28042,0.155789,0.436209,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.115935
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LINE:9 [9,53469,1883,tcp,mqtt,32.048637,9,5,3,3,0.280023,0.156013,0.436836,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.11593
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LINE:10 [10,54153,1883,tcp,mqtt,31.977057,9,5,3,3,0.281452,0.156362,0.437814,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.115
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LINE:11 [11,39671,1883,tcp,mqtt,31.962308,9,5,3,3,0.281582,0.156434,0.438016,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.75,0.8.333333,12.980
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LINE:13 [13,51495,1883,tcp,mqtt,31.885127,9,5,3,3,0.282263,0.156813,0.439076,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.444444,13.115
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LINE:15 [15,36349,1883,tcp,mqtt,32.0614159999999994,9,5,3,3,0.280711,0.155951,0.436662,0.555556,296,32,40,168,32,40,0,2,1,3,3,13,0,0,0,0,0.33,0.76,0.8.4444
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4444,13.719559999999999,0,23,32,0,0,4,9.555183,0,0,33,0,188,0,7,714286,11.618477,324,222107,29746104,081999,32047288.177939,4805511.622245,1068143.1670  
01,1923.799515,1518199.9935,20412668.06648,510517,001152,682383.534838,77.089201,29746104,081999,32047288.177939,246516.0138,8204056.271833,3.37062,3,0,1.6  
66667,25.333333,0,0,666667,0,0,0,0,0,0,0,0,0,0,0,0,2301184.177399,231184.177399,2301184.177399,2301184.177399,0,0,29746104,081999,29746104.081999,  
29746104.081999,0,64240,26847,502,MQTT_Publish]
```

```

TESTING PARTITIONING          4
PAIR CONTAINERf partitions:
38667, 32.0116
51143, 31.8836
44761, 32.1241
60893, 31.9611
51087, 31.9024
48579, 31.8697
54063, 32.0947
33457, 32.104
52181, 32.027
53469, 32.0486
54153, 31.9771
39671, 31.9623
44225, 31.9653
51495, 31.8851
42037, 31.9266
36349, 32.0614
39763, 32.0251
57501, 31.9082
56117, 32.0092
44927, 31.9305
36375, 32.0486
42381, 32.0543
58171, 31.927
39989, 32.0384
54593, 31.9695
52085, 31.9502
41525, 32.0473
37425, 31.9116

ANALYZING FLOWS
Total flow: 881sec

C:\Users\admin\OneDrive - Concordia University - Canada\coen assignment 4 ii\x64\Debug\coen assignment 4 ii.exe (process 28984) exited with code 0 (0x0).
Press any key to close this window . . .|

```

The 7 created files from partitions=4

 shard_0.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_1.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_2.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_3.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_4.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_5.txt		4/13/2026 12:15 AM	Text Document	4 KB
 shard_6.txt		4/13/2026 12:15 AM	Text Document	4 KB